Declarations

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Programme	B.Tech Computer Science and Engineering
Project	AUTOSAR HLD Document Analysis Assistant (Case Study 1)

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1. Academic integrity declaration

I declare that:

- This submission is my original work, except where stated below.
- All external code, datasets, models and documents used are cited below.
- The results reported have not been fabricated; they can be reproduced with `Code/autosar_hld_assistant/evaluation/run_evaluation.py` and the test suite.
- This is an individual submission, and my contribution is stated accurately below.

My contribution

- Selected Case Study 1 and defined the system: Streamlit UI, local embeddings, FAISS, Mistral LLM, SQLite storage, human-in-the-loop review.
- Built the first working version of the application (ingestion, entity extraction including the table parsers, grounded chat, review tab, export).
- Decided the fixes and features for the final version, tested the application, ran the evaluation.
- Recorded and narrated the demo video.

External sources used

Item	Type	Licence / source
Automotive Engineering AI Project Case Studies, Case Study 1	Problem specification	Course material
Vehicle Brake Control System HLD (VBCHLD-001 rev 1.0, marked "Synthetic Test Document")	Main input / test document	Obtained from an external website
AUTOSAR EXP Virtual Functional Bus; CP SWS Platform Types; TR Foundation Release Overview	Reference documents	© AUTOSAR GbR, autosar.org
all-MiniLM-L6-v2 (sentence-transformers)	Pre-trained embedding model	Apache-2.0, HuggingFace
Mistral `ministral-8b-latest`	Pre-trained LLM via API	Mistral AI terms of service
Streamlit, FAISS, sentence-transformers, pypdf, OpenAI Python SDK, FastAPI, pandas, NumPy, python-dotenv, ReportLab, pytest	Libraries	Open-source (Apache-2.0 / MIT / BSD)

Signature: R.S.KIRAN SOORYA Date: 13-10-2026

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2. AI-tool usage declaration

AI model / tool	Purpose	Artifact affected	Verified by student
Claude Code (Anthropic, Claude Opus 5.5)	Coding and documentation assistant, working on my existing application	Refactoring and bug fixes in my original modules (`app/core`, `app/ingestion`, `app/extraction`, `app/rag`, `app/storage`, `app/reports`); UI redesign (`app/ui`); added modules `app/analysis`, `app/api.py`, `app/pipeline.py`; unit tests (`tests/`); evaluation script; sample revision 1.3; the files in `Documentation/` and `Model_Prompts_Config/`; the synopsis (`Synopsis/`); the technical report (`126003133_Report_v1.0.pdf`) and its screenshots (`Documentation/126003133_ReportFigures_v1.0/`); the narration script used for the demo video	Ran the full unit-test suite (84 tests, all passing); ran the evaluation script with and without the language model and checked the results against the documents; used every page of the application with the sample documents; checked answers against the cited PDF pages; read each module and can explain how it works
Mistral `ministral-8b-latest`	LLM inside the application: generates grounded answers; refines entity extraction	Runtime answers only (no code)	Evaluated with `run_evaluation.py --llm`; answers checked against cited pages
all-MiniLM-L6-v2	Embedding model inside the application	Vector index (runtime)	Retrieval evaluated in `Evaluation_Results/`

No other AI tools were used.

I remain accountable for the correctness of the submitted work and can explain all of it.

Signature: R.S.KIRAN SOORYA Date: 13-10-2026